

CSE420: Compiler Design
Self Study Assignment 03

Submit with a cover page containing your Name, ID,
Section (MUST)
Deadline: 8th Jan

- Given the SDT, evaluate the following string with either annotated parse tree or dependency graph

String: **while (a > b) c = d + e; if (x > y) u = w + v; else u = p + q**

SDT:

PRODUCTION	SEMANTIC RULES
$P \rightarrow S$	$S.next = newlabel()$ $P.code = S.code \parallel label(S.next)$
$S \rightarrow \text{if} (B) S_1 \text{ else } S_2$	$B.true = newlabel()$ $B.false = newlabel()$ $S_1.next = S_2.next = S.next$ $S.code = B.code$ $\parallel label(B.true) \parallel S_1.code$ $\parallel gen('goto' S.next)$ $\parallel label(B.false) \parallel S_2.code$
$S \rightarrow \text{while} (B) S_1$	$begin = newlabel()$ $B.true = newlabel()$ $B.false = S.next$ $S_1.next = begin$ $S.code = label(begin) \parallel B.code$ $\parallel label(B.true) \parallel S_1.code$ $\parallel gen('goto' begin)$
$S \rightarrow \text{id} = E ;$	$S.code = E.code \parallel$ $gen(top.get(\text{id.lexeme}) '=' E.addr)$
$E \rightarrow E_1 + E_2$	$E.addr = new Temp()$ $E.code = E_1.code \parallel E_2.code \parallel$ $gen(E.addr '=' E_1.addr '+' E_2.addr)$
$\mid \text{id}$	$E.addr = top.get(\text{id.lexeme})$ $E.code = ''$
$B \rightarrow E_1 \text{ rel } E_2$	$B.code = E_1.code \parallel E_2.code$ $\parallel gen('if' E_1.addr \text{ rel.op } E_2.addr 'goto' B.true)$ $\parallel gen('goto' B.false)$

Note that for final assignment the question is missing the rule $S \rightarrow S_1 S_2$. You need to incorporate that to solve.